

Chapter 3: Linear Transformation, Eigenvalues and Eigenvectors

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3.1 Linear Transformation

In this chapter, we are going to look at matrices from a different perspective by treating them as some transformations. More precisely, we shall see how a matrix transform a vector to another vector through matrix multiplication. This provides a geometrical interpretation for matrices in the lower dimensional cases.

3.1.1 Matrix as a Mapping

We start with a simple 2×2 matrix $\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

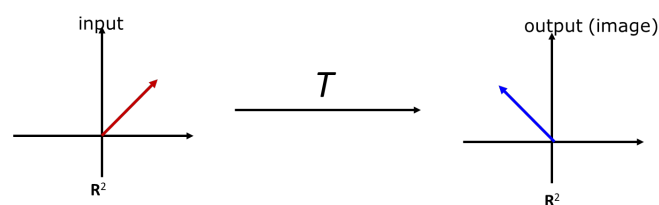
Given this 2-vector $\mathbf{u} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ which we treat it as an input vector, we shall process it under matrix $\mathbf{A}$ by means of multiplication.

The resulting product $\mathbf{A}\mathbf{u}$ is another 2-vector which we call it the output vector, or the image.

$$\mathbf{u} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \xrightarrow{\text{red arrow}} \mathbf{A} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \xrightarrow{\text{red arrow}} \mathbf{A}\mathbf{u} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

In other words, we can think of a matrix as a mapping from $\mathbf{R}^2$ to $\mathbf{R}^2$.

Geometrically, this mapping transforms the arrow representing the vector $\mathbf{u}$ to this arrow representing the vector $\mathbf{A}\mathbf{u}$:



Here we denote the mapping by the letter T , which stands for transformation.

Notation:

$T : \mathbf{R}^2 \rightarrow \mathbf{R}^2 \text{ defined by } T(\mathbf{u}) = \mathbf{A}\mathbf{u} \text{ for all } \mathbf{u} \text{ in } \mathbf{R}^2$

We shall call such a mapping a linear transformation.

3.1.2 Linear Transformation

We now look at a general input vector $\mathbf{u} = \begin{pmatrix} x \\ y \end{pmatrix}$

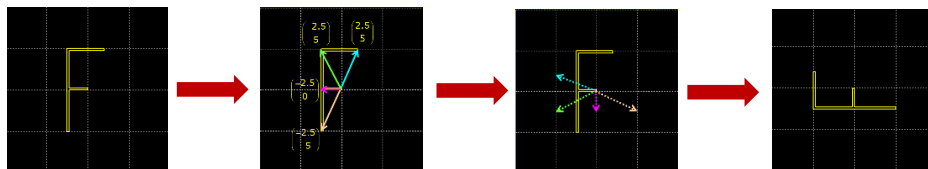
We process this input vector like before and get the output vector $\mathbf{Au}$:

$$\underset{\text{input}}{\mathbf{u} = \begin{pmatrix} x \\ y \end{pmatrix}} \xrightarrow{\quad} \mathbf{A} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \xrightarrow{\quad} \underset{\text{output (image)}}{\mathbf{Au} = \begin{pmatrix} -y \\ x \end{pmatrix}}$$

This gives us a formula of the linear transformation:

$$T: \mathbf{R}^2 \rightarrow \mathbf{R}^2 \text{ given by } T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} -y \\ x \end{pmatrix} \text{ for all } \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbf{R}^2$$

The geometrical interpretation of this matrix can be seen more clearly by considering a slightly more complicated diagram, like the letter F below.



We see that the resulting diagram is the letter of F “lying down”. In fact, the effect of this transformation is the 90° counterclockwise rotation about the origin. In fact, many 2 x 2 matrices give transformations that have interesting geometrical interpretation on the xy-plane.

3.1.3 m x n Linear Transformation

Now we look at a general m x n matrix $\mathbf{A}$ and go through the same procedure. In order to perform the matrix multiplication, the input vector $\mathbf{u}$ must be an n-vector and the output vector $\mathbf{Au}$ will be an m-vector.

$$\underset{\text{input}}{\mathbf{u} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}} \xrightarrow{\quad} \mathbf{A} = m \times n \text{ matrix} \xrightarrow{\quad} \underset{\text{output (image)}}{\mathbf{Au} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{pmatrix}}$$

So this gives a mapping T with domain $\mathbf{R}^n$ and codomain $\mathbf{R}^m$.

T is called a linear transformation from $\mathbf{R}^n$ to $\mathbf{R}^m$ with notation

$$T: \mathbf{R}^n \rightarrow \mathbf{R}^m \text{ defined by } T(\mathbf{u}) = \mathbf{Au} \text{ for all } \mathbf{u} \in \mathbf{R}^n.$$

$\mathbf{A}$ is called the standard matrix of the linear transformation.

$$\text{Suppose } \mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} \text{ and } \mathbf{u} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}.$$

This gives the formula for T:

$$T\left(\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}\right) = \begin{pmatrix} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n \end{pmatrix}$$

3.1.4 Linear Transformation by Formula

Supposed we are given a mapping in terms of formula as follow:

$$T: \mathbf{R}^2 \rightarrow \mathbf{R}^3 \text{ such that } T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x+y \\ 2x \\ -3y \end{pmatrix}$$

We can regard x and y as two parameters, and rewrite the formula as a linear combination as shown in the middle term below:

$$T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x+y \\ 2x \\ -3y \end{pmatrix} = x\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + y\begin{pmatrix} 1 \\ 0 \\ -3 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 2 & 0 \\ 0 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

Then this linear combination can be rewritten in matrix form with $\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 0 \\ -3 \end{pmatrix}$ forming the two columns of a 3 x 2 matrix $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 2 & 0 \\ 0 & -3 \end{pmatrix}$.

So we see that T is a linear transformation with this given standard matrix $\mathbf{A}$.

3.1.5 Images of Standard Basis

Let $T: \mathbf{R}^n \rightarrow \mathbf{R}^m$ be any linear transformation with the standard matrix given by

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}.$$

Suppose we use the standard basis vectors of $\mathbf{R}^n$

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, \mathbf{e}_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

as the input vectors of this mapping. Then

$$T(\mathbf{e}_1) = \mathbf{A}\mathbf{e}_1 = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{pmatrix}$$

We notice that this output vector is precisely the first column of the standard matrix $\mathbf{A}$.

Similarly, if we take the k^{th} standard basis vector as the input vectors, the output vectors will correspond to the k^{th} column of $\mathbf{A}$. In other words,

the image $T(\mathbf{e}_k)$ = the k^{th} column of $\mathbf{A}$ and the matrix $\mathbf{A} = (T(\mathbf{e}_1) \ T(\mathbf{e}_2) \ \dots \ T(\mathbf{e}_n))$.

3.1.6 Properties of Linear Transformation

If $T : \mathbf{R}^n \rightarrow \mathbf{R}^m$ is a linear transformation, then

1. $T(\mathbf{0}) = \mathbf{0}$ (T preserves zero vector)
2. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ (T preserves addition)
3. $T(c\mathbf{u}) = cT(\mathbf{u})$ (T preserves scalar multiplication)
4. $T(c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \dots + c_k\mathbf{u}_k) = c_1T(\mathbf{u}_1) + c_2T(\mathbf{u}_2) + \dots + c_kT(\mathbf{u}_k)$ (T preserves linear combinations)

All these properties are referred to as the linearity properties of the linear transformation, and they can be verified easily by converting the map T into matrix multiplication by the standard matrix $\mathbf{A}$ to the input vector. For example, in the last property, the left hand side can be written as pre-multiplying $\mathbf{A}$ to the linear combination:

$$\begin{aligned} T(c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \dots + c_k\mathbf{u}_k) &= \mathbf{A}(c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \dots + c_k\mathbf{u}_k) \\ &= c_1\mathbf{A}\mathbf{u}_1 + c_2\mathbf{A}\mathbf{u}_2 + \dots + c_k\mathbf{A}\mathbf{u}_k \\ &= c_1T(\mathbf{u}_1) + c_2T(\mathbf{u}_2) + \dots + c_kT(\mathbf{u}_k) \end{aligned}$$

where the second equality is a property of matrix multiplication.

Example: Linearity Property

Given a linear transformation $T : \mathbf{R}^2 \rightarrow \mathbf{R}^2$, with both the standard matrix and formula unknown. The only information we have is that the images of two particular input vectors

$$T\left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} 4 \\ 2 \end{pmatrix} \quad T\left(\begin{pmatrix} 1 \\ -1 \end{pmatrix}\right) = \begin{pmatrix} 0 \\ 6 \end{pmatrix}$$

Based on this piece of information, how do we find the images of $\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ under T ?

We shall see that the linearity property of T can help us.

First observe that the two input vectors form a basis for $\mathbf{R}^2$ (just need to check they are linearly independent). This means it is possible to express $\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ as linear combinations of this basis.

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{1}{2}\begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (\text{use Gaussian elimination, or simply by inspection})$$

$$\begin{aligned} T\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) &= T\left(\frac{1}{2}\begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{1}{2}\begin{pmatrix} 1 \\ -1 \end{pmatrix}\right) = \frac{1}{2}T\left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right) + \frac{1}{2}T\left(\begin{pmatrix} 1 \\ -1 \end{pmatrix}\right) \quad (\text{by linearity property}) \\ &= \frac{1}{2}\begin{pmatrix} 4 \\ 2 \end{pmatrix} + \frac{1}{2}\begin{pmatrix} 0 \\ 6 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix} \quad (\text{from given information above}) \end{aligned}$$

Likewise,

$$\begin{aligned} \begin{pmatrix} 0 \\ 1 \end{pmatrix} &= \frac{1}{2}\begin{pmatrix} 1 \\ 1 \end{pmatrix} - \frac{1}{2}\begin{pmatrix} 1 \\ -1 \end{pmatrix} \\ T\left(\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right) &= T\left(\frac{1}{2}\begin{pmatrix} 1 \\ 1 \end{pmatrix} - \frac{1}{2}\begin{pmatrix} 1 \\ -1 \end{pmatrix}\right) = \frac{1}{2}T\left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right) - \frac{1}{2}T\left(\begin{pmatrix} 1 \\ -1 \end{pmatrix}\right) = \frac{1}{2}\begin{pmatrix} 4 \\ 2 \end{pmatrix} - \frac{1}{2}\begin{pmatrix} 0 \\ 6 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \end{pmatrix} \end{aligned}$$

Recall from an earlier slide, the columns of the standard matrix $\mathbf{A}$ of the linear transformation is given by the image of the standard basis $\mathbf{e}_1$ and $\mathbf{e}_2$, we can therefore recover:

$$\text{Standard Matrix } \mathbf{A} = (T(\mathbf{e}_1) \ T(\mathbf{e}_2)) = \begin{pmatrix} 2 & 2 \\ 4 & -2 \end{pmatrix}.$$

3.1.7 Stacking Method

In the previous example, we see that when we are given the images of a basis for $\mathbf{R}^2$ under the linear transformation T , we can find the standard matrix A of T by first finding the images of the standard basis e_1 and e_2 . In fact, there is a shortcut method that allows us to find the matrix A directly. This is known as the “stacking” method.

We shall illustrate using the same linear transformation $T : \mathbf{R}^2 \rightarrow \mathbf{R}^2$ above.

From the given information $T\left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$ and $T\left(\begin{pmatrix} 1 \\ -1 \end{pmatrix}\right) = \begin{pmatrix} 0 \\ 6 \end{pmatrix}$

we can replace T by A : $A\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$ and $A\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 6 \end{pmatrix}$ (*)

Matrix properties allow us to combine the two equations in (*) into one:

$$A\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 2 & 6 \end{pmatrix}$$

Observe that we have stacked the two input vectors (in blue) into the blue matrix on the left, and stacked the two output vectors (in green) into the green matrix on the right. Since the two column vectors of the blue matrix are linearly independent, this matrix is non-singular. So we can compute:

$$A = \begin{pmatrix} 4 & 0 \\ 2 & 6 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}^{-1} = \begin{pmatrix} 2 & 2 \\ 4 & -2 \end{pmatrix}.$$

This stacking method can be extended to any linear transformation with domain $\mathbf{R}^n$ where the images of a basis for $\mathbf{R}^n$ is given.

3.1.8 Images of a basis

Given linear transformation $T : \mathbf{R}^n \rightarrow \mathbf{R}^m$ and a basis $\{u_1, u_2, \dots, u_n\}$ for $\mathbf{R}^n$.

For any v in $\mathbf{R}^n$, we have $v = c_1u_1 + c_2u_2 + \dots + c_nu_n$ and hence

$$T(v) = c_1T(u_1) + c_2T(u_2) + \dots + c_nT(u_n)$$

Hence, knowing the images $T(u_1), T(u_2), \dots, T(u_n)$ is enough to determine:

- the image $T(v)$ of any vector v in the domain $\mathbf{R}^n$;
- the standard matrix of T ;
- the formula of T .

We say:

the linear transformation T is completely determined by the images $T(u_1), T(u_2), \dots, T(u_n)$ of the basis $\{u_1, u_2, \dots, u_n\}$.

3.2 Eigenvalues and Eigenvectors

Given a square matrix, we can talk about the eigenvalues and eigenvectors associated to this matrix. It is probably the most commonly used concepts in linear algebra that is being applied to solving Engineering problems. In chapter 4, we shall see how these concepts are being used in solving system of differential equations.

3.2.1 Eigenvalues and Eigenvectors

Let $\mathbf{A}$ be $n \times n$ square matrix. By treating $\mathbf{A}$ as a linear transformation, we have

$$\mathbf{A} \begin{bmatrix} \text{input} \end{bmatrix} = \begin{bmatrix} \text{output} \end{bmatrix}$$

Let $\mathbf{x}$ be a nonzero (input) vector in $\mathbf{R}^n$. If the output vector $\mathbf{Ax}$ is a scalar multiple of $\mathbf{x}$, then we call $\mathbf{x}$ an eigenvector of $\mathbf{A}$.

If we write $\mathbf{Ax} = \lambda \mathbf{x}$, then the scalar λ (*pronounced: "lambda"*) is called an eigenvalue of $\mathbf{A}$; and $\mathbf{x}$ is said to be an eigenvector of $\mathbf{A}$ associated with the eigenvalue λ .

Geometrically (in 2- or 3-spaces), the arrows representing $\mathbf{x}$ and $\mathbf{Ax}$ are parallel, pointing either in the same or opposite directions.

Example 1:

$$\mathbf{A} = \begin{pmatrix} 0.96 & 0.01 \\ 0.04 & 0.99 \end{pmatrix}$$

Let's use the following as the input vectors:

$$\mathbf{x} = \begin{pmatrix} 1 \\ 4 \end{pmatrix} \quad \mathbf{y} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Then

$$\mathbf{Ax} = \begin{pmatrix} 0.96 & 0.01 \\ 0.04 & 0.99 \end{pmatrix} \begin{pmatrix} 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \end{pmatrix} = \mathbf{x}$$

So $\mathbf{x}$ is an eigenvector of $\mathbf{A}$ with the eigenvalue 1.

And

$$\mathbf{Ay} = \begin{pmatrix} 0.96 & 0.01 \\ 0.04 & 0.99 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0.95 \\ -0.95 \end{pmatrix} = 0.95 \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 0.95\mathbf{y}$$

So $\mathbf{y}$ is an eigenvector of $\mathbf{A}$ with the eigenvalue 0.95.

Example 2:

We shall look at another matrix $\mathbf{B}$ with input vectors $\mathbf{x}, \mathbf{y}, \mathbf{z}$:

$$\mathbf{B} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \quad \mathbf{x} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad \mathbf{y} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \quad \mathbf{z} = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$$

First, we have

$$\mathbf{B}\mathbf{x} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \\ 3 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 3\mathbf{x}$$

So $\mathbf{x}$ is an eigenvector associated with eigenvalue 3.

Before we move on to $\mathbf{y}$ and $\mathbf{z}$, let us look at the scalar multiples of $\mathbf{x}$:

$$\mathbf{B}(2\mathbf{x}) = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 6 \\ 6 \\ 6 \end{pmatrix} = 3 \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix} = 3(2\mathbf{x})$$

So $2\mathbf{x}$ is also an eigenvector associated with eigenvalue 3.

In fact, for any non-zero scalar k , $k\mathbf{x}$ is an eigenvector associated with eigenvalue 3:

$$\mathbf{B}(k\mathbf{x}) = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} k \\ k \\ k \end{pmatrix} = 3 \begin{pmatrix} k \\ k \\ k \end{pmatrix} = 3(k\mathbf{x})$$

In general,

if $\mathbf{x}$ is an eigenvector associated with eigenvalue λ ,
any scalar multiples $k\mathbf{x}$ will also be an eigenvector associated with the same eigenvalue λ .

Now for $\mathbf{y}$ and $\mathbf{z}$,

$$\mathbf{B}\mathbf{y} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = 0 \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = 0\mathbf{y}$$

$$\mathbf{B}\mathbf{z} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = 0 \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} = 0\mathbf{z}$$

So $\mathbf{y}$ and $\mathbf{z}$ are both eigenvectors associated with eigenvalue 0.

Note:

1. 0 can be an eigenvalue, but the zero vector $\mathbf{0}$ cannot be an eigenvector.
2. Two eigenvectors that are not scalar multiples of each other may have the same eigenvalue.

3.2.2 Finding Eigenvalues without Eigenvectors

Without the eigenvectors, how do we find eigenvalues of a matrix? The answer is by solving a certain polynomial. We shall demonstrate with an example below.

$$\mathbf{A} = \begin{pmatrix} 0.96 & 0.01 \\ 0.04 & 0.99 \end{pmatrix}$$

$$\det(\lambda\mathbf{I} - \mathbf{A}) = \det\left(\lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 0.96 & 0.01 \\ 0.04 & 0.99 \end{pmatrix}\right) = \begin{vmatrix} \lambda - 0.96 & -0.01 \\ -0.04 & \lambda - 0.99 \end{vmatrix} = (\lambda - 0.96)(\lambda - 0.99) - (-0.01)(-0.04)$$

So we have $\det(\lambda\mathbf{I} - \mathbf{A}) = \lambda^2 - 1.95\lambda + 0.95$, which is a polynomial in λ of degree 2.

To find the eigenvalues of $\mathbf{A}$, we solve the polynomial equation:

$$\lambda^2 - 1.95\lambda + 0.95 = 0$$

which gives the solutions $\lambda = 1$ and 0.95 , and these give all the eigenvalues of $\mathbf{A}$.

3.2.3 Characteristic Polynomial

Let $\mathbf{A}$ be $n \times n$ square matrix. Then $\det(\lambda \mathbf{I} - \mathbf{A})$ is a polynomial of degree n . This polynomial is called the characteristic polynomial of $\mathbf{A}$.

Example: The characteristic polynomial of $\mathbf{A} = \begin{pmatrix} 0.96 & 0.01 \\ 0.04 & 0.99 \end{pmatrix}$ is $\lambda^2 - 1.95\lambda + 0.95$

We have:

$$\lambda \text{ is an eigenvalue of } \mathbf{A} \Leftrightarrow \det(\lambda \mathbf{I} - \mathbf{A}) = 0 \Leftrightarrow \lambda \text{ is a root of the characteristic polynomial}$$

We shall now explain why this works.

We will need to recall a few concepts that we have learned in earlier chapters.

First of all, by definition, we have

λ is an eigenvalue of $\mathbf{A} \Leftrightarrow \mathbf{A}\mathbf{x} = \lambda\mathbf{x}$ for some nonzero column vector $\mathbf{x}$

$$\Leftrightarrow \lambda\mathbf{x} - \mathbf{A}\mathbf{x} = \mathbf{0} \quad \text{for some nonzero column vector } \mathbf{x}$$

We can further rewrite $\lambda\mathbf{x} - \mathbf{A}\mathbf{x} = \mathbf{0}$ as $(\lambda \mathbf{I} - \mathbf{A})\mathbf{x} = \mathbf{0}$.

So the condition now become:

$$(\lambda \mathbf{I} - \mathbf{A})\mathbf{x} = \mathbf{0} \quad (*) \quad \text{for some nonzero column vector } \mathbf{x} \quad (\Delta)$$

The matrix equation $(*)$ with $\mathbf{x}$ as unknown represents a homogeneous system with the coefficient matrix $\lambda \mathbf{I} - \mathbf{A}$.

Condition (Δ) implies this linear system has a non-trivial solution. Now recall that a homogeneous system has non-trivial solution means the coefficient matrix $\lambda \mathbf{I} - \mathbf{A}$ is singular, and

$$\det(\lambda \mathbf{I} - \mathbf{A}) = 0 \quad (\dagger)$$

Now if we expand the determinant using the inductive expansion (see section 1.7), the equation $(\dagger)$ will turn into a polynomial equation in terms of λ . In order for us to find the eigenvalue λ , we need to solve this polynomial equation.

Example:

$$\mathbf{C} = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & 2 \\ 1 & 1 & 1 \end{pmatrix}$$

To find the eigenvalues of $\mathbf{C}$, we solve its characteristic polynomial:

$$\begin{aligned} \det(\lambda \mathbf{I} - \mathbf{C}) &= \begin{vmatrix} \lambda & 1 & 0 \\ 0 & \lambda & -2 \\ -1 & -1 & \lambda-1 \end{vmatrix} = \lambda \begin{vmatrix} \lambda & -2 \\ -1 & \lambda-1 \end{vmatrix} - 1 \begin{vmatrix} 0 & -2 \\ -1 & \lambda-1 \end{vmatrix} + 0 \begin{vmatrix} 0 & \lambda \\ -1 & -1 \end{vmatrix} \quad (\text{using inductive expansion}) \\ &= \lambda^3 - \lambda^2 - 2\lambda + 2 \end{aligned}$$

It may not be easy to factorize a degree 3 polynomial. A short cut is to try guessing one possible root. In this case, it is not difficult to guess the root $\lambda = 1$. Ultimately, the complete factorization of this polynomial is given by

$$\det(\lambda \mathbf{I} - \mathbf{C}) = (\lambda - 1)(\lambda^2 - 2) = (\lambda - 1)(\lambda - \sqrt{2})(\lambda + \sqrt{2})$$

The eigenvalues of $\mathbf{C}$ are the roots of the above polynomial, which are 1, $\sqrt{2}$ and $-\sqrt{2}$.

3.2.4 Triangular Matrices

If $\mathbf{A}$ is a triangular matrix, the eigenvalues of $\mathbf{A}$ are all the diagonal entries of $\mathbf{A}$.
In particular, the eigenvalues of a diagonal matrix are its diagonal entries.

For example,

$\begin{pmatrix} -1 & 3.5 & 14 \\ 0 & 5 & -26 \\ 0 & 0 & 2 \end{pmatrix}$ This is an upper triangular matrix with diagonal entries -1 , 5 and 2 .
So the eigenvalues are -1 , 5 and 2 .

$\begin{pmatrix} -2 & 0 & 0 \\ 99 & 0 & 0 \\ 10 & -4.5 & 10 \end{pmatrix}$ This is a lower triangular matrix with diagonal entries -2 , 0 and 10 .
So the eigenvalues are -2 , 0 and 10 .

3.2.5 Eigenvalue and Singularity

If 0 is an eigenvalue of $\mathbf{A}$, then $\mathbf{A}$ is singular. Otherwise, $\mathbf{A}$ is non-singular.

Below is an explanation:

- 0 is an eigenvalue of $\mathbf{A}$
- $\Leftrightarrow 0$ is a root of the characteristic polynomial $\det(\lambda \mathbf{I} - \mathbf{A})$
- $\Leftrightarrow \det(0\mathbf{I} - \mathbf{A}) = 0$
- $\Leftrightarrow \det(-\mathbf{A}) = 0$
- $\Leftrightarrow \det(\mathbf{A}) = 0$
- $\Leftrightarrow \mathbf{A}$ is singular

Note that the arguments go both ways.

3.2.6 Matrix with Complex Eigenvalues

To end this section, we make a remark about complex eigenvalues.

Given a matrix $\mathbf{A}$, it is possible that the characteristic polynomial $\det(\lambda \mathbf{I} - \mathbf{A})$ may have complex roots.

Example $\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

The characteristic polynomial of this matrix is $\det(\lambda \mathbf{I} - \mathbf{A}) = \lambda^2 + 1$. It has complex roots: $\lambda = \pm i$.

So this matrix has complex eigenvalues. The corresponding eigenvectors will have complex components. We shall see some applications of complex eigenvalues and eigenvectors in solving system of differential equations in the last chapter.

3.3 Eigenspaces

We have seen in the previous section that, given an $n \times n$ square matrix, there is a systematic way to find the eigenvalues by solving the characteristic polynomial. In this section, we shall see how to find the eigenvectors associated to a certain eigenvalue. The collection of all such eigenvectors is called an eigenspace, which is a subspace of $\mathbb{R}^n$.

3.3.1 Eigenspace

Given an $n \times n$ square matrix $\mathbf{A}$.

Suppose λ is an eigenvalue of $\mathbf{A}$. Then $\det(\lambda \mathbf{I} - \mathbf{A}) = 0$. This means the homogeneous linear system

$$(\lambda \mathbf{I} - \mathbf{A}) \mathbf{x} = \mathbf{0}$$

with coefficient matrix $\lambda \mathbf{I} - \mathbf{A}$ has non-trivial solutions. In fact,

all the non-trivial solutions of the system $(\lambda \mathbf{I} - \mathbf{A}) \mathbf{x} = \mathbf{0}$ are the eigenvectors of $\mathbf{A}$ associated to λ .

The solution space of this homogeneous system is called the eigenspace of $\mathbf{A}$ associated with λ and we denote this subspace of $\mathbb{R}^n$ by E_λ .

If $\mathbf{u}$ is a nonzero vector in E_λ , then $\mathbf{u}$ is an eigenvector of $\mathbf{A}$ associated with the eigenvalue λ .

Example 1:

$$\mathbf{A} = \begin{pmatrix} 0.96 & 0.01 \\ 0.04 & 0.99 \end{pmatrix}$$

In the previous section, we have found the eigenvalues of $\mathbf{A}$ to be 1 and 0.95 by solving its characteristic polynomial $\det(\lambda \mathbf{I} - \mathbf{A}) = (\lambda - 1)(\lambda - 0.95)$. So correspondingly there are two eigenspace E_1 and $E_{0.95}$ associated to eigenvalues 1 and 0.95 respectively. We solve for these two eigenspaces separately.

For $\lambda = 1$, the homogeneous system $(\lambda \mathbf{I} - \mathbf{A}) \mathbf{x} = \mathbf{0}$ is given by

$$(\mathbf{I} - \mathbf{A}) \mathbf{x} = \mathbf{0} \quad \Leftrightarrow \quad \begin{pmatrix} 1 - 0.96 & -0.01 \\ -0.04 & 1 - 0.99 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Using Gaussian elimination, we can solve the system and get the general solution

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0.25t \\ t \end{pmatrix} = t \begin{pmatrix} 0.25 \\ 1 \end{pmatrix}$$

with parameter t . We can pull out the parameter t as a scalar as shown on the right side above. This means any non-zero scalar multiple of $\begin{pmatrix} 0.25 \\ 1 \end{pmatrix}$ is an eigenvector of $\mathbf{A}$ associated with the eigenvalue 1. The eigenspace is given by

$$E_1 = \text{span} \left\{ \begin{pmatrix} 0.25 \\ 1 \end{pmatrix} \right\}$$

We can also take this eigenvector $\begin{pmatrix} 0.25 \\ 1 \end{pmatrix}$ (or any one of its multiples) as a basis for E_1 .

We do the same for $\lambda = 0.95$.

$$(0.95\mathbf{I} - \mathbf{A}) \mathbf{x} = \mathbf{0} \quad \Leftrightarrow \quad \begin{pmatrix} 0.95 - 0.96 & -0.01 \\ -0.04 & 0.95 - 0.99 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Again using Gaussian elimination, we can solve the system and get the general solution

$$\begin{pmatrix} x \\ y \end{pmatrix} = t \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

Any non-zero scalar multiple of $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$ is an eigenvector of $\mathbf{A}$ associated with the eigenvalue 0.95.

The eigenspace $E_{0.95}$ is given by

$$E_{0.95} = \text{span}\left\{\begin{pmatrix} -1 \\ 1 \end{pmatrix}\right\}$$

and a basis for $E_{0.95}$ is given by $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$ or any of its multiples.

Example 2:

$$\mathbf{B} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

We have seen before that the eigenvalues of $\mathbf{B}$ are 3 and 0 by solving the characteristic polynomial $\det(\lambda \mathbf{I} - \mathbf{B}) = (\lambda - 3)\lambda^2$. So $\mathbf{B}$ has two eigenspaces E_3 and E_0 associated to eigenvalues 0 and 3 respectively.

These eigenspaces can be solved in a similar way. Here we will only find the eigenspace E_0 .

We substitute $\lambda = 0$ in the homogeneous system $(\lambda \mathbf{I} - \mathbf{B}) \mathbf{x} = 0$ to get

$$(-\mathbf{B}) \mathbf{x} = 0 \quad \Leftrightarrow \quad \begin{pmatrix} 0 & -1 & -1 & -1 \\ -1 & 0 & -1 & -1 \\ -1 & -1 & 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Using Gaussian elimination, we solve the system to get the general solution with two parameters s and t :

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = s \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

This means any non-zero linear combination of $\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$ is an eigenvector of $\mathbf{B}$ associated with the eigenvalue 0. The eigenspace can be expressed as the linear span of these two vectors:

$$E_0 = \text{span}\left\{\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}\right\}$$

and these two vectors form a basis for the eigenspace E_0 .

3.3.2 Eigenspace of Identity Matrix

We use the 3 x 3 identity matrix as an example, but the result can be generalized to any $n \times n$ identity matrix.

$$\mathbf{I}_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

We have the following properties:

1. The identity matrix I_3 has only one eigenvalue 1.
2. I_3 has only one eigenspace E_1 .
3. The eigenspace E_1 is $\mathbf{R}^3$.

Property 1 is easy to see, as I_3 is a diagonal matrix, and eigenvalues of a diagonal matrix are all its diagonal entries, which is just 1 in this case. In fact, if you consider the characteristic polynomial of the identity matrix, it is given by $\det(\lambda I_3 - I_3) = (\lambda - 1)^3$, which has only one (repeated) root 1.

Property 2 follows from 1 immediately.

Property 3 says that any non-zero 3-vector in $\mathbf{R}^3$ is an eigenvector. This is straight forward by definition, since for any vector $\mathbf{v}$ in $\mathbf{R}^3$,

$$I_3 \mathbf{v} = \mathbf{v}.$$

Alternatively, we can look at the matrix

$$\lambda I_3 - I_3 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

with $\lambda = 1$, which turns out to be the zero matrix. Then

$$(\lambda I_3 - I_3) \mathbf{x} = \mathbf{0} \implies \mathbf{0x} = \mathbf{0}$$

This homogeneous system with zero matrix as the coefficient is satisfied by every vector in $\mathbf{R}^3$. This shows that the eigenspace E_1 is $\mathbf{R}^3$.

3.3.3 Multiplicity of Eigenvalues

For the three matrices we have seen in this section, let us recap their characteristic polynomials.

$$\mathbf{A} = \begin{pmatrix} 0.96 & 0.01 \\ 0.04 & 0.99 \end{pmatrix}$$

The characteristic polynomial of $\mathbf{A}$ is given by this factorized form $\det(\lambda I - \mathbf{A}) = (\lambda - 1)(\lambda - 0.95)$.

Note that the power of each of the two factors $(\lambda - 1)$ and $(\lambda - 0.95)$ is 1. We call these powers the respective multiplicities of the eigenvalues.

$$\mathbf{B} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

The characteristic polynomial of $\mathbf{B}$ is $\det(\lambda I - \mathbf{B}) = (\lambda - 3) \lambda^2$

The factor $(\lambda - 3)$ corresponding to eigenvalue 3 has power 1, and the factor λ^2 corresponding to eigenvalue 0 has power 2. So we say the eigenvalue 3 has multiplicity 1, and the eigenvalue 0 has multiplicity 2.

$$\mathbf{I}_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

The characteristic polynomial of $\mathbf{I}$ is $\det(\lambda I - \mathbf{I}) = (\lambda - 1)^3$ has only one factor with power 3. So the eigenvalue 1 has multiplicity 3.

Now let's look at the dimension of the various eigenspaces. Recall that the dimension is the number of basis vectors for a certain vector space.

For matrix $\mathbf{A}$, we have $\dim E_1 = 1$, $\dim E_{0.95} = 1$.

For matrix $\mathbf{B}$, we have $\dim E_3 = 1$, $\dim E_0 = 2$.

For the identity matrix $\mathbf{I}$, $\dim E_1 = \dim \mathbf{R}^3 = 3$.

Observe that the dimension of the eigenspace is equal to the multiplicity of the corresponding eigenvalues given above.

This is not true in general. However, there is a relationship between the two quantities.

3.3.4 Dimension of Eigenspace

Suppose the characteristic polynomial of a matrix $\mathbf{A}$ is factorized as follow, with all the common factors group together:

$$\det(\lambda \mathbf{I} - \mathbf{A}) = (\lambda - \lambda_1)^{r_1} (\lambda - \lambda_2)^{r_2} \cdots (\lambda - \lambda_k)^{r_k}$$

What we know from this polynomial are: λ_1 to λ_k are all the eigenvalues of $\mathbf{A}$, r_1 to r_k are the respective multiplicities of these eigenvalues. If $\mathbf{A}$ is an $n \times n$ matrix, then

$$\begin{aligned} r_1 + r_2 + \dots + r_k &= n \\ \dim E_{\lambda_i} &\leq r_i \text{ for all } i \end{aligned}$$

The inequality above says that the number of basis vectors in each eigenspace cannot be more than the multiplicity of the eigenvalue in the characteristic polynomial.

Here's an example.

Suppose $\det(\lambda \mathbf{I} - \mathbf{A}) = (\lambda - 1)^1 (\lambda - 2)^3 (\lambda - 4)^2$ is a characteristic polynomial of a 6×6 matrix $\mathbf{A}$. So we know there are three eigenvalues 1, 2 and 4.

What is the dimension of each of their eigenspaces?

Let's look at E_2 first. Since the multiplicity of eigenvalue 2 is equal to 3, we know

$$\dim E_2 \leq 3.$$

In other words, the dimension can be 1, 2 or 3, but we can't tell the exact value until we find the eigenspace explicitly.

Similarly, for E_4 , the multiplicity of the eigenvalue 4 in the characteristic polynomial is 2, so

$$\dim E_4 \leq 2.$$

As for E_1 , since the multiplicity of eigenvalue 1 in the characteristic polynomial is 1, we conclude that

$$\dim E_1 = 1$$

because dimension of an eigenspace cannot be anything less than 1.

3.4 Diagonalizable Matrices

Diagonal matrix is a special type of matrix that has many nice features. For matrices that are not diagonal, we would like to “convert” them to diagonal matrices. Those matrices that can be converted are known as diagonalizable matrices. In this section, we will give a more precise description, and show how this notion is related to eigenvalues and eigenvectors.

3.4.1 Power of Matrix

We start with an example as a motivation, which is about taking powers of square matrices.

$$\mathbf{A} = \begin{pmatrix} 0.96 & 0.01 \\ 0.04 & 0.99 \end{pmatrix}$$

We want to compute $\mathbf{A}^n = \begin{pmatrix} 0.96 & 0.01 \\ 0.04 & 0.99 \end{pmatrix}^n$

We cannot simply raise each entry to power n , because of the way we perform matrix multiplication.

What we will do is to perform a “factorization” on the matrix $\mathbf{A}$ in the following way:

$$\mathbf{A} = \begin{pmatrix} 0.96 & 0.01 \\ 0.04 & 0.99 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 1 \\ 4 & -1 \end{pmatrix}}_{\mathbf{P}} \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 0.95 \end{pmatrix}}_{\mathbf{D}} \underbrace{\begin{pmatrix} 1 & 1 \\ 4 & -1 \end{pmatrix}^{-1}}_{\mathbf{P}^{-1}}$$

The right hand side is the product of three matrices, which will be explained later how they are derived. (You may try to work backward to verify that the product is indeed equal to $\mathbf{A}$.)

The two outer matrices in the product are inverses of each other, which we denote by $\mathbf{P}$ and $\mathbf{P}^{-1}$ and the matrix in the middle is a diagonal matrix, which we denote by $\mathbf{D}$. So we have

$$\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$$

Our objective is to raise $\mathbf{A}$ to the power n . So we do the same to the product on the right.

$$\mathbf{A}^n = (\mathbf{P}\mathbf{D}\mathbf{P}^{-1})^n = (\mathbf{P}\mathbf{D}\mathbf{P}^{-1})(\mathbf{P}\mathbf{D}\mathbf{P}^{-1})(\mathbf{P}\mathbf{D}\mathbf{P}^{-1}) \dots (\mathbf{P}\mathbf{D}\mathbf{P}^{-1}) \quad (n \text{ times}) \quad (*)$$

(Note that $(\mathbf{P}\mathbf{D}\mathbf{P}^{-1})^n \neq \mathbf{P}^n \mathbf{D}^n \mathbf{P}^{-n}$)

One nice property about matrix multiplication is the associative law, which allows us to rearrange the parenthesis on the right hand side of $(*)$:

$$\mathbf{P}\mathbf{D}(\mathbf{P}^{-1}\mathbf{P})\mathbf{D}(\mathbf{P}^{-1}\mathbf{P})\mathbf{D}(\mathbf{P}^{-1}\mathbf{P}) \dots (\mathbf{P}^{-1}\mathbf{P})\mathbf{D}\mathbf{P}^{-1} \quad (**)$$

We observe that there are many pairs of $\mathbf{P}^{-1}\mathbf{P}$ side by side in $(**)$. All the intermediate pairs of $\mathbf{P}^{-1}\mathbf{P}$ can be cancelled as their product is the identity matrix and we are left with a much simpler product:

$$\mathbf{A}^n = \mathbf{P}\mathbf{D}\mathbf{D} \dots \mathbf{D}\mathbf{P}^{-1} = \mathbf{P}\mathbf{D}^n\mathbf{P}^{-1}.$$

Note that on the right hand side, we raise $\mathbf{D}$ to the power of n , while the powers of $\mathbf{P}$ and $\mathbf{P}^{-1}$ remain unchanged.

As mentioned, diagonal matrices have many nice features. One of them being easy to perform multiplication. In particular, raising a diagonal matrix to power n , is just raising its individual diagonal entries to power n . So for our $\mathbf{D}$, we have

$$\mathbf{D}^n = \begin{pmatrix} 1 & 0 \\ 0 & 0.95 \end{pmatrix}^n = \begin{pmatrix} 1^n & 0 \\ 0 & 0.95^n \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0.95^n \end{pmatrix}$$

For example, if we want to compute $\mathbf{A}^{100}$, instead of multiplying matrix $\mathbf{A}$ with itself 100 times, we express it as:

$$\begin{aligned}\mathbf{A}^{100} &= \mathbf{P}\mathbf{D}^{100}\mathbf{P}^{-1} \\ &= \begin{pmatrix} 1 & 1 \\ 4 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0.95^{100} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 4 & -1 \end{pmatrix}^{-1} \\ &= \begin{pmatrix} 0.2047 & 0.1988 \\ 0.7953 & 0.8012 \end{pmatrix}\end{aligned}$$

Essentially we just need to perform multiplication on three matrices as shown. And this is a more efficient way to get the answer.

3.4.2 Diagonalizable Matrix

From our previous example, we have the factorization of $\mathbf{A}$ given by

$$\begin{pmatrix} 0.96 & 0.01 \\ 0.04 & 0.99 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 4 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0.95 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 4 & -1 \end{pmatrix}^{-1}$$

which is represented algebraically by $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$. We can rewrite this equality as:

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{D}$$

which in the expanded form is given by:

$$\begin{pmatrix} 1 & 1 \\ 4 & -1 \end{pmatrix}^{-1} \begin{pmatrix} 0.96 & 0.01 \\ 0.04 & 0.99 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 4 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0.95 \end{pmatrix}$$

What we have done is converting a **non-diagonal matrix** $\mathbf{A}$ to a **diagonal matrix** $\mathbf{D}$. When such a conversion can be carried out, we say the matrix $\mathbf{A}$ is a diagonalizable matrix, and $\mathbf{P}$ is the matrix that diagonalizes $\mathbf{A}$ to give us a diagonal matrix $\mathbf{D}$.

Let's state the definition more generally:

An $n \times n$ square matrix $\mathbf{A}$ is called diagonalizable if we can find a non-singular matrix $\mathbf{P}$ such that $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ is a diagonal matrix:

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{pmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \ddots \\ & & & \lambda_n \end{pmatrix}$$

We say: the matrix $\mathbf{P}$ diagonalizes $\mathbf{A}$. Matrix $\mathbf{P}$ is necessarily non-singular, as we need to take its inverse matrix.

Example:

Here's another example of 3×3 matrix $\mathbf{B}$ and the corresponding matrix $\mathbf{P}$:

$$\mathbf{B} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \quad \mathbf{P} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & -2 \\ 1 & -1 & 1 \end{pmatrix}$$

We can check that $\mathbf{P}^{-1}\mathbf{B}\mathbf{P}$ is a diagonal matrix!

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & -2 \\ 1 & -1 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & -2 \\ 1 & -1 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$\mathbf{P}^{-1} \quad \mathbf{B} \quad \mathbf{P}$

So $\mathbf{B}$ is a diagonalizable matrix, and $\mathbf{P}$ diagonalizes $\mathbf{B}$.

How do we get this matrix $\mathbf{P}$? We shall see at the end of this section that $\mathbf{P}$ is related to the eigenvectors of $\mathbf{B}$.

3.4.3 Non-diagonalizable Matrix

Not all square matrices are diagonalizable. Let us look at this example $\mathbf{M}$, which is a non-diagonalizable matrix.

$$\mathbf{M} = \begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix}$$

In other words, we cannot find a matrix $\mathbf{P}$ that diagonalizes $\mathbf{M}$. We illustrate below why this is the case by “brute force” (see remark Δ below)

Suppose, on the contrary, there is a $\mathbf{P}$ such that $\mathbf{P}^{-1}\mathbf{M}\mathbf{P}$ is diagonal.

Let's write this non-singular matrix as $\mathbf{P} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

Then $\det(\mathbf{P}) = ad - bc \neq 0$ (*).

Now let us expand the product $\mathbf{P}^{-1}\mathbf{M}\mathbf{P}$ and equate it with some diagonal matrix $\begin{pmatrix} e & 0 \\ 0 & f \end{pmatrix}$.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} \begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} e & 0 \\ 0 & f \end{pmatrix}$$

Then

$$\frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \begin{pmatrix} 2a & 2b \\ a+2c & b+2d \end{pmatrix} = \begin{pmatrix} e & 0 \\ 0 & f \end{pmatrix}$$

$$\frac{1}{ad-bc} \begin{pmatrix} 2ad-b(a+2c) & 2bd-b(b+2d) \\ -2ac+a(a+2c) & -2bc+a(b+2d) \end{pmatrix} = \begin{pmatrix} e & 0 \\ 0 & f \end{pmatrix}$$

The next thing is to compare the entries on both sides of the equality above.

For the (1,2)-entry, we have $b^2 = 0$ which implies $b = 0$.

For the (2,1)-entry, we have $a^2 = 0$ which implies $a = 0$.

This will imply: $\det(\mathbf{P}) = ad - bc = 0$ which contradict (*) above.

This means the above assumption is wrong, which implies that we cannot find the required matrix $\mathbf{P}$.

(Δ) The above is an ad hoc method to show a non-diagonalizable matrix. In the next section, we will see a more systematic way to determine whether a matrix is diagonalizable or not.

3.4.4 Diagonalizability

Let us make a comparison of the three matrices:

$$\mathbf{A} = \begin{pmatrix} 0.96 & 0.01 \\ 0.04 & 0.99 \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \quad \mathbf{M} = \begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix}$$

We have seen that $\mathbf{A}$, $\mathbf{B}$ are diagonalizable, while $\mathbf{M}$ is not diagonalizable.

Let us now bring the eigenvalues and eigenvectors into the discussion.

$\mathbf{A}$ has two eigenvalues 1 and 0.95, $\mathbf{B}$ also has two eigenvalues 3 and 0, while $\mathbf{M}$ has one eigenvalue 2 (observe that $\mathbf{M}$ is a triangular matrix).

However, it is not the number of eigenvalues that decide whether the matrix is diagonalizable or not. It is the eigenvectors.

For matrix $\mathbf{A}$, the two eigenvalues each correspond to an eigenvector: $\begin{pmatrix} 1 \\ 4 \end{pmatrix}$ $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ (note that these are scalar multiples of the eigenvectors $\begin{pmatrix} 0.25 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$ respectively on page 10.

For matrix **B**, we have one eigenvector associated to eigenvalue 3, $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ and two eigenvectors associated to eigenvalue 0: $\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$ $\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$

For matrix **M**, I leave it to you to check that an eigenvector of **M** associated to eigenvalue 2 is given by $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Note that we are not saying the matrices above have only 2, 3 and 1 eigenvector respective. In fact, these matrices have infinitely many eigenvectors.

For example, for **M**, any scalar multiple of $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ is also an eigenvector: $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ $\begin{pmatrix} 0 \\ 2 \end{pmatrix}$ $\begin{pmatrix} 0 \\ 3 \end{pmatrix}$...

The correct way to say this is to include the condition linearly independent:

In other words, matrix **A** has two linearly independent eigenvectors; matrix **B** has three linearly independent eigenvectors; while **M**, it has only one linearly independent eigenvector.

In the last case, there are “not enough” linearly independent eigenvectors. But how do we tell whether there are enough eigenvectors for a given matrix? The answer is to compare it with the size of the matrix.

3.4.5 Diagonalizability and Eigenvectors

The following result gives the condition for a matrix to be diagonalizable.

Let **A** be an $n \times n$ square matrix.

If **A** has n linearly independent eigenvectors, then **A** is diagonalizable

If we cannot find n eigenvectors of **A** that are linearly independent, then **A** is not diagonalizable.

(We will give an explanation of this result in 3.4.7.)

Using the examples above:

A is 2×2 and has 2 linearly independent eigenvectors. So **A** is diagonalizable.

B is 3×3 and has 3 linearly independent eigenvectors. So **B** is diagonalizable.

M is 2×2 and only has 1 linearly independent eigenvector. So **M** is non-diagonalizable.

3.4.6 Two Useful Observations

The following two observations, not only explain the condition for diagonalizability, but they are also techniques used commonly in matrix multiplications.

Observation 1: Matrix multiplication of two matrices **A** and **B**.

If we write matrix **B** in terms of its columns $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n$. Then

$$\mathbf{AB} = \mathbf{A}(\mathbf{b}_1 \ \mathbf{b}_2 \ \dots \ \mathbf{b}_n) = (\mathbf{Ab}_1 \ \mathbf{Ab}_2 \ \dots \ \mathbf{Ab}_n)$$

In other words, the 1st column of **AB** is given by $\mathbf{Ab}_1$, the 2nd column of **AB** is given by $\mathbf{Ab}_2$ and so on.

To illustrate, let us look at this concrete example:

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} \mathbf{b}_1 & \mathbf{b}_2 & \mathbf{b}_3 \end{pmatrix} = \begin{pmatrix} \mathbf{Ab}_1 & \mathbf{Ab}_2 & \mathbf{Ab}_3 \end{pmatrix}$$

You can first multiply the two matrices on the left to see that you will get the right hand side.

Now denote the three columns of B as b_1, b_2, b_3 .

Then you can check that the first column of AB is Ab_1 , the second column is Ab_2 , and the third column is Ab_3 .

Observation 2: Matrix multiplication of two matrices B and D , where D is a diagonal matrix.

Suppose the diagonal entries of D are $d_1, d_2, \dots, d_n$, and the columns of B are $b_1, b_2, \dots, b_n$. Then

$$BD = (b_1 \ b_2 \ \dots \ b_n) D = (d_1 b_1 \ d_2 b_2 \ \dots \ d_n b_n)$$

Let us use an example:

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 5 \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{pmatrix} = \begin{pmatrix} 2 & 3 & 4 \\ 2 & 6 & 12 \\ 2 & 9 & 20 \end{pmatrix}$$

$b_1 \ b_2 \ b_3 \qquad D \qquad 2b_1 \ 3b_2 \ 4b_3$

The product of the two matrices on the left hand side yield the matrix on the right.

Let b_1, b_2, b_3 be the three columns of the matrix on the left, we again check the matrix on the right by columns, and see that column 1 is $2b_1$, column 2 is $3b_2$, and column 3 is $4b_3$.

3.4.7 Diagonalizability and Eigenvectors – proof [Optional reading]

We want to show: if an $n \times n$ matrix A has n linearly independent eigenvectors, then it is diagonalizable.

First, we denote the n eigenvectors by $u_1, u_2, \dots, u_n$ and let $\lambda_1, \lambda_2, \dots, \lambda_n$ be the corresponding eigenvalues. (Note that some of these n eigenvalues may be the same.)

Then we form the $n \times n$ matrix P by using the n eigenvectors as its columns and let D be the diagonal matrix where the entries are the eigenvalues:

$$P = (u_1 \ u_2 \ \dots \ u_n) \text{ and } D = \begin{pmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \lambda_n \end{pmatrix}$$

Note that P is guaranteed to be non-singular because the columns are linearly independent.

$$\text{Now } AP = (Au_1 \ Au_2 \ \dots \ Au_n) = (\lambda_1 u_1 \ \lambda_2 u_2 \ \dots \ \lambda_n u_n)$$

The first equality above follows from the first observation in the previous segment.

The second equality is due to $u_1, u_2, \dots, u_n$ being eigenvectors of A , so that $Au_i = \lambda_i u_i$.

Now from the second observation in the previous segment,

$$(\lambda_1 u_1 \ \lambda_2 u_2 \ \dots \ \lambda_n u_n) = (u_1 \ u_2 \ \dots \ u_n) \begin{pmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \lambda_n \end{pmatrix}$$

The product on the right is precisely PD .

So putting everything together, we get $AP = PD$.

And by bringing the P on the right hand side to the left, we get $P^{-1}AP = D$.

This means the matrix P diagonalizes A to give the diagonal matrix D and so we conclude that A is diagonalizable.

3.5 Diagonalization

In this section, we will introduce a systematic procedure to determine whether a matrix is diagonalizable or not. The approach will also show how to diagonalize a matrix. This process is called diagonalization.

3.5.1 Algorithm for Diagonalization

Step 1: Solve the characteristic polynomial $\det(\lambda I - \mathbf{A})$ to find all distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_k$.

Step 2: For each λ_i , find a basis S_{λ_i} for the eigenspace E_{λ_i} by solving $(\lambda_i I - \mathbf{A})\mathbf{x} = \mathbf{0}$

Step 3: Let $S = S_{\lambda_1} \cup S_{\lambda_2} \cup \dots \cup S_{\lambda_k}$. (Then $|S|$ is the total number of basis vectors from all the eigenspaces)

(a) If $|S| < n$, then $\mathbf{A}$ is not diagonalizable.

In this case, there are not enough linearly independent eigenvectors to form a matrix $\mathbf{P}$ that diagonalizes $\mathbf{A}$.

(b) If $|S| = n$, then $\mathbf{A}$ is diagonalizable.

In this case, if $S = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$, then the square matrix $\mathbf{P} = (\mathbf{u}_1 \ \mathbf{u}_2 \ \dots \ \mathbf{u}_n)$ diagonalizes $\mathbf{A}$.

Example 1: Diagonalizable Matrix

$$\mathbf{B} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

Step 1: By solving the characteristic polynomial, the eigenvalues are 3 and 0.

Step 2: For $\lambda = 3$, solve $(3I - \mathbf{B})\mathbf{x} = \mathbf{0}$; for $\lambda = 0$, solve $(0I - \mathbf{B})\mathbf{x} = \mathbf{0}$

We have $S_3 = \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}$ a basis for E_3 and $S_0 = \left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\}$ a basis for E_0

Step 3: $|S| = |S_3| + |S_0| = 3$, which agree with the size 3×3 of matrix $\mathbf{B}$.

So we conclude $\mathbf{B}$ is diagonalizable.

We form the matrix $\mathbf{P}$ using the three vectors in S as columns: $\mathbf{P} = \begin{pmatrix} 1 & -1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$

$$\text{Then } \mathbf{P}^{-1}\mathbf{BP} = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

The diagonal matrix on the right has the eigenvalues of $\mathbf{B}$ as the diagonal entries. Note that 0 appears two times as there are two eigenvectors associated to this eigenvalue.

Note: In step 3, it is not necessary for you to actually perform the multiplication $\mathbf{P}^{-1}\mathbf{BP}$. All you need to do is to write down the diagonal matrix using the eigenvalues found in step 1.

Remark: Matrix $\mathbf{P}$ is not unique. There are many possible matrices that can diagonalize $\mathbf{B}$ to give a diagonal matrix. We can use any other eigenvectors of $\mathbf{B}$ as the columns of $\mathbf{P}$, as long as they are linearly independent.

For example, $\mathbf{P} = \begin{pmatrix} 1 & -1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$ can be replaced by $\mathbf{P} = \begin{pmatrix} 2 & -7 & 1 \\ 2 & 7 & 0 \\ 2 & 0 & -1 \end{pmatrix}$

$\mathbf{u}_1 \quad \mathbf{u}_2 \quad \mathbf{u}_3$ $2\mathbf{u}_1 \quad 7\mathbf{u}_2 \quad -\mathbf{u}_3$

Note that the three columns in the matrix on the right are scalar multiples of the original eigenvectors, and hence are still eigenvectors of $\mathbf{B}$.

We can also re-order the columns.

For example, matrix $\mathbf{Q}$ is obtained from $\mathbf{P}$ by moving the first column to the last. As long as the three columns are linearly independent eigenvectors, it will still diagonalize the matrix.

$$\mathbf{Q} = \begin{pmatrix} -1 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

$\mathbf{u}_2 \quad \mathbf{u}_3 \quad \mathbf{u}_1$

You can check that $\mathbf{Q}^{-1}\mathbf{BQ} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{pmatrix}$

Take note that the order of the diagonal entries on the right is rearranged accordingly.

Example 2: Non-Diagonalizable Matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ -3 & 5 & 2 \end{pmatrix}$$

Step 1: The eigenvalues are 1 and 2. ($\mathbf{A}$ is a lower triangular matrix, so the eigenvalues can be read off from the diagonal entries directly.)

Step 2: For $\lambda = 1$, solve $(\mathbf{I} - \mathbf{A})\mathbf{x} = 0$; for $\lambda = 2$, solve $(2\mathbf{I} - \mathbf{A})\mathbf{x} = 0$.

We have a basis $\mathbf{S}_1 = \left\{ \begin{pmatrix} 1 \\ -1 \\ 8 \end{pmatrix} \right\}$ for E_1 and a basis $\mathbf{S}_2 = \left\{ \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$ for E_2

Step 3: $|\mathbf{S}| = |\mathbf{S}_1| + |\mathbf{S}_2| = 2$.

Since we only have two linearly independent eigenvectors, so $\mathbf{A}$ is not diagonalizable.

3.5.2 Dimension of Eigenspace

Recall from section 3.3, the characteristic polynomial of an $n \times n$ matrix $\mathbf{A}$:

$$\det(\lambda \mathbf{I} - \mathbf{A}) = (\lambda - \lambda_1)^{r_1} (\lambda - \lambda_2)^{r_2} \cdots (\lambda - \lambda_k)^{r_k}$$

where $r_1, r_2, \dots, r_k$ are called the multiplicities of the eigenvalues of $\mathbf{A}$. And we have

$$r_1 + r_2 + \dots + r_k = n \quad \text{and} \quad \dim E_{\lambda_i} \leq r_i \text{ for all } i.$$

Note that $\dim E_{\lambda_i} = |\mathbf{S}_{\lambda_i}|$ in the algorithm above.

Case 1: If $\dim E_{\lambda_i} = r_i$ for all i , then $|\mathbf{S}| = r_1 + r_2 + \dots + r_k = n$.

In this case, $\mathbf{A}$ is diagonalizable.

Case 2: If $\dim E_{\lambda_i} < r_i$ for at least one i , then $|\mathbf{S}| < r_1 + r_2 + \dots + r_k = n$.

In this case, $\mathbf{A}$ is not diagonalizable.

3.5.3 Matrix with Maximum Distinct Eigenvalues

Let $\mathbf{A}$ be an $n \times n$ square matrix.

If $\mathbf{A}$ has n distinct eigenvalues, then $\mathbf{A}$ is diagonalizable.

Suppose the n eigenvalues of $\mathbf{A}$ are $\lambda_1, \lambda_2, \dots, \lambda_n$. Then each eigenvalue will correspond to an eigenvector $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n$. Since all the eigenvalues are distinct, the eigenvectors will be linearly independent. So we have n linearly independent eigenvectors for $\mathbf{A}$. This implies $\mathbf{A}$ is diagonalizable.

What this observation tells us is that, when we carry out the diagonalization algorithm, after step 1, if we obtain n distinct eigenvalues, then we can straight away conclude that the matrix is diagonalizable without going through the remaining steps.

However, if we need to find the matrix $\mathbf{P}$ that diagonalizes $\mathbf{A}$, we still need to perform steps 2 and 3 of the algorithm to find explicit eigenvectors.

Example:

$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 2 & 2 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 4 \end{pmatrix}$ Observe that this is an upper triangular matrix with diagonal entries 1, 2, 3, 4. So we know immediately $\mathbf{A}$ has 4 distinct eigenvalues and hence $\mathbf{A}$ is a diagonalizable matrix.

$\mathbf{B} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}$ This is a diagonal matrix with diagonal entries 1, 1, 2, 2. So $\mathbf{B}$ has only 2 distinct eigenvalues. Is $\mathbf{B}$ diagonalizable? All diagonal matrices are automatically diagonalizable.

So $\mathbf{B}$ is also a diagonalizable matrix despite not having a maximum number of distinct eigenvalues.

Hence, if a matrix is an $n \times n$ diagonalizable matrix, it need not have n distinct eigenvalues.

3.6 Powers of Matrices

In this last section, we shall revisit the power of square matrices that are diagonalizable. We will also give an example on application of power of matrices to population modeling.

3.6.1 Iterative Systems

Many real life examples come in the form of iterative systems.

These are systems with various stages (usually over a period of time) such that:

- the stages can be described using n-vectors, $\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2, \dots$
- consecutive stages are related by a fixed $n \times n$ matrix $\mathbf{A}$: $\mathbf{x}_k = \mathbf{A}\mathbf{x}_{k-1}$

where $\mathbf{x}_{k-1}$ and $\mathbf{x}_k$ are n-vectors representing the $(k-1)^{\text{th}}$ and k^{th} stage respectively.

Here is an example of a system where the stages are represented by 3-vectors:

$$\begin{array}{ccccccc} \begin{pmatrix} x_0 \\ y_0 \\ z_0 \end{pmatrix} & \rightarrow & \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} & \rightarrow & \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix} & \cdots & \rightarrow & \begin{pmatrix} x_n \\ y_n \\ z_n \end{pmatrix} \\ \text{stage 0} & & \text{stage 1} & & \text{stage 2} & & & \text{stage n} \\ \text{(initial)} & & & & & & & \\ \mathbf{x}_0 & & \mathbf{x}_1 = \mathbf{A}\mathbf{x}_0 & & \mathbf{x}_2 = \mathbf{A}\mathbf{x}_1 & & & \mathbf{x}_n = \mathbf{A}\mathbf{x}_{n-1} \end{array}$$

The matrix $\mathbf{A}$ in this case is a 3×3 matrix.

The stage 1 vector is given by $\mathbf{x}_1 = \mathbf{A}\mathbf{x}_0$.

The stage 2 vector is given by $\mathbf{x}_2 = \mathbf{A}\mathbf{x}_1 = \mathbf{A}(\mathbf{A}\mathbf{x}_0) = \mathbf{A}^2\mathbf{x}_0$.

In a similar manner, the stage n vector is given by

$$\mathbf{x}_n = \mathbf{A}\mathbf{x}_{n-1} = \mathbf{A}(\mathbf{A}\mathbf{x}_{n-2}) = \mathbf{A}(\mathbf{A}(\mathbf{A}\mathbf{x}_{n-3})) = \dots = \mathbf{A}^n\mathbf{x}_0$$

3.6.2 Power of Diagonalizable Matrices

Let $\mathbf{A}$ be a diagonalizable matrix of order n and $\mathbf{P}$ a non-singular matrix such that

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{pmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \ddots \\ & & & \lambda_n \end{pmatrix}$$

Raising both sides to power m :

$$(\mathbf{P}^{-1}\mathbf{A}\mathbf{P})^m = \begin{pmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \ddots \\ & & & \lambda_n \end{pmatrix}^m$$

The left side can be simplified as $\mathbf{P}^{-1}\mathbf{A}^m\mathbf{P}$

while the right hand side is given by this matrix

$$\begin{pmatrix} \lambda_1^m & & 0 \\ & \lambda_2^m & \\ 0 & & \ddots \\ & & & \lambda_n^m \end{pmatrix}$$

So we have $\mathbf{P}^{-1}\mathbf{A}^m\mathbf{P} = \begin{pmatrix} \lambda_1^m & & 0 \\ & \lambda_2^m & \\ 0 & & \ddots \\ & & & \lambda_n^m \end{pmatrix}$

$$\text{Then } \mathbf{A}^m = \mathbf{P} \begin{pmatrix} \lambda_1^m & & 0 \\ & \lambda_2^m & \\ 0 & & \ddots \\ & & & \lambda_n^m \end{pmatrix} \mathbf{P}^{-1}$$

This gives a simple way to compute and analyse large powers of $\mathbf{A}$ by just performing matrix multiplication on three matrices.

Example:

$$\mathbf{A} = \begin{pmatrix} -4 & 0 & -6 \\ 2 & 1 & 2 \\ 3 & 0 & 5 \end{pmatrix}$$

By going through the algorithm for diagonalization, we obtain (details skipped):

$$\mathbf{P} = \begin{pmatrix} -2 & 0 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \quad \text{and} \quad \mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$$\text{Then } \mathbf{A}^m = \mathbf{P} \begin{pmatrix} (-1)^m & 0 & 0 \\ 0 & 1^m & 0 \\ 0 & 0 & 2^m \end{pmatrix} \mathbf{P}^{-1}$$

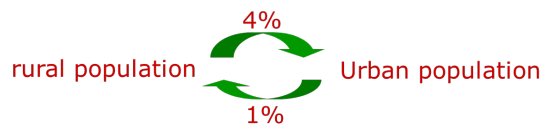
In fact, since this matrix is non-singular, we can even apply negative powers to this. In particular, we can let m be -1 to get $\mathbf{A}^{-1}$ in the same form.

$$\mathbf{A}^{-1} = \mathbf{P} \begin{pmatrix} (-1)^{-1} & 0 & 0 \\ 0 & 1^{-1} & 0 \\ 0 & 0 & 2^{-1} \end{pmatrix} \mathbf{P}^{-1}$$

3.6.3 Population Modeling

We now apply the power of matrix to population modeling.

Suppose the population in a certain country is divided into rural and urban. Studies show that every year, 4% of rural population will move to the urban area, while 1% of the urban population will move to the rural area.



Suppose the current rural and urban populations are 40,000 and 60,000 respectively.

- What will be the projected rural and urban populations after n years?
- What will be the projected proportion of the two populations in the long term?

As this is an iterative system, we need to analyse the population in terms of the year. So we let

a_n = the rural population after n years, and b_n = the urban population after n years, with respect to current year as the starting point.

So a_0 = the current rural population = 40,000 and b_0 = the current rural population = 60,000.

Now based on the information we have, the rural population after n years depends on both the rural and urban populations after $n-1$ years:

$$a_n = 0.96a_{n-1} + 0.01b_{n-1}$$

In other words, 96% of rural population after $n-1$ years and 1% of urban population after $n-1$ year will contribute to the rural population after n years.

Similarly, 4% of rural population after $n-1$ years and 99% of urban population after $n-1$ years will contribute to the urban population after n years.

$$b_n = 0.04a_{n-1} + 0.99b_{n-1}$$

This is a kind of linear system that we can convert into the matrix equation form as before.

$$\begin{aligned} a_n &= 0.96a_{n-1} + 0.01b_{n-1} \\ b_n &= 0.04a_{n-1} + 0.99b_{n-1} \end{aligned} \rightarrow \begin{pmatrix} a_n \\ b_n \end{pmatrix} = \begin{pmatrix} 0.96 & 0.01 \\ 0.04 & 0.99 \end{pmatrix} \begin{pmatrix} a_{n-1} \\ b_{n-1} \end{pmatrix}$$

Let $\mathbf{A} = \begin{pmatrix} 0.96 & 0.01 \\ 0.04 & 0.99 \end{pmatrix}$ and $\mathbf{x}_n = \begin{pmatrix} a_n \\ b_n \end{pmatrix}$. Then we have the iterative relation

$$\mathbf{x}_n = \mathbf{A}\mathbf{x}_{n-1} = \mathbf{A}(\mathbf{A}\mathbf{x}_{n-2}) = \mathbf{A}(\mathbf{A}(\mathbf{A}\mathbf{x}_{n-3})) = \dots = \mathbf{A}^n\mathbf{x}_0$$

as before. Here $\mathbf{x}_0 = \begin{pmatrix} a_0 \\ b_0 \end{pmatrix} = \begin{pmatrix} 40,000 \\ 60,000 \end{pmatrix}$ represents the current population.

To find the population after n years, we will look at a_n and b_n , which means we need to find the n^{th} stage vector $\mathbf{x}_n$. And from what we derive, we should be looking at $\mathbf{A}^n$. Using diagonalization, we have

$$\mathbf{A}^n = \begin{pmatrix} 1 & 1 \\ 4 & -1 \end{pmatrix} \begin{pmatrix} 1^n & 0 \\ 0 & 0.95^n \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 4 & -1 \end{pmatrix}^{-1}$$

The diagonal entries are 1 and 0.95, which are the eigenvalues of $\mathbf{A}$. So

$$\mathbf{x}_n = \mathbf{A}^n\mathbf{x}_0 = \begin{pmatrix} 1 & 1 \\ 4 & -1 \end{pmatrix} \begin{pmatrix} 1^n & 0 \\ 0 & 0.95^n \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 4 & -1 \end{pmatrix}^{-1} \begin{pmatrix} 40,000 \\ 60,000 \end{pmatrix}$$

After simplifying the right hand side, we get

$$\mathbf{x}_n = \begin{pmatrix} a_n \\ b_n \end{pmatrix} = \begin{pmatrix} \frac{1}{5}(1+4(0.95)^n) & \frac{1}{5}(1-(0.95)^n) \\ \frac{1}{5}(4-4(0.95)^n) & \frac{1}{5}(4+(0.95)^n) \end{pmatrix} \begin{pmatrix} 40,000 \\ 60,000 \end{pmatrix} = \begin{pmatrix} 20,000(1+0.95^n) \\ 20,000(4-0.95^n) \end{pmatrix}$$

To determine the long term population, we will find the n^{th} stage vector for large n .

And from what we derive, we should be looking at $\mathbf{A}^n$ for some large power n of $\mathbf{A}$. Again we have

$$\mathbf{A}^{(\text{big } n)} = \begin{pmatrix} 1 & 1 \\ 4 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0.95^{(\text{big } n)} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 4 & -1 \end{pmatrix}^{-1}$$

Since $0.95 < 1$, when it is raised to a large power, it will be closed to 0. So we can approximate this entry with 0 and hence the resulting product can be computed to be approximately given by this:

$$\mathbf{A}^{(\text{big } n)} \approx \begin{pmatrix} 1 & 1 \\ 4 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 4 & -1 \end{pmatrix}^{-1} = \begin{pmatrix} 0.2 & 0.2 \\ 0.8 & 0.8 \end{pmatrix}$$

We can then use this matrix to find the populations in the long term:

$$\mathbf{x}_{(\text{big } n)} = \begin{pmatrix} a_{(\text{big } n)} \\ b_{(\text{big } n)} \end{pmatrix} = \begin{pmatrix} 0.2 & 0.2 \\ 0.8 & 0.8 \end{pmatrix} \begin{pmatrix} 40,000 \\ 60,000 \end{pmatrix} = \begin{pmatrix} 0.2(100,000) \\ 0.8(100,000) \end{pmatrix}$$

The resulting vector shows that the rural population in the long term is 20% of the total population, and the urban population in the long term is 80% of the total population.

Of course, this is a very simple model of the population, with many assumptions made. For example, factors such as migration, birth and death are not taken into account.